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The Hydro-Emissions Tension: Addressing Structural Vulnerabilities in Canada's Net-Zero Transition

Summary

Canada's electricity grid is quite clean due to a heavy reliance on hydro generation, but recent droughts have started to highlight a structural risk: when hydro generation drops, the system often falls back on fossil generation or imports to fill the shortfall. This is already visible in provinces like British Columbia and Manitoba, and it risks becoming more pronounced as electricity demand rises with electrification. The issue is not hydro itself, but the limited availability of reliable, non-emitting backup capacity and the fragmented structure of the national grid. Without expanding other clean energy sources, improving interprovincial transmission, and revising how grid planners account for hydro's weather-dependence, even short-term reductions in hydro output can lead to higher emissions if not offset by other clean sources. Over time, this could leave Canada structurally more dependent on fossil fuels during periods of supply pressure.

Introduction

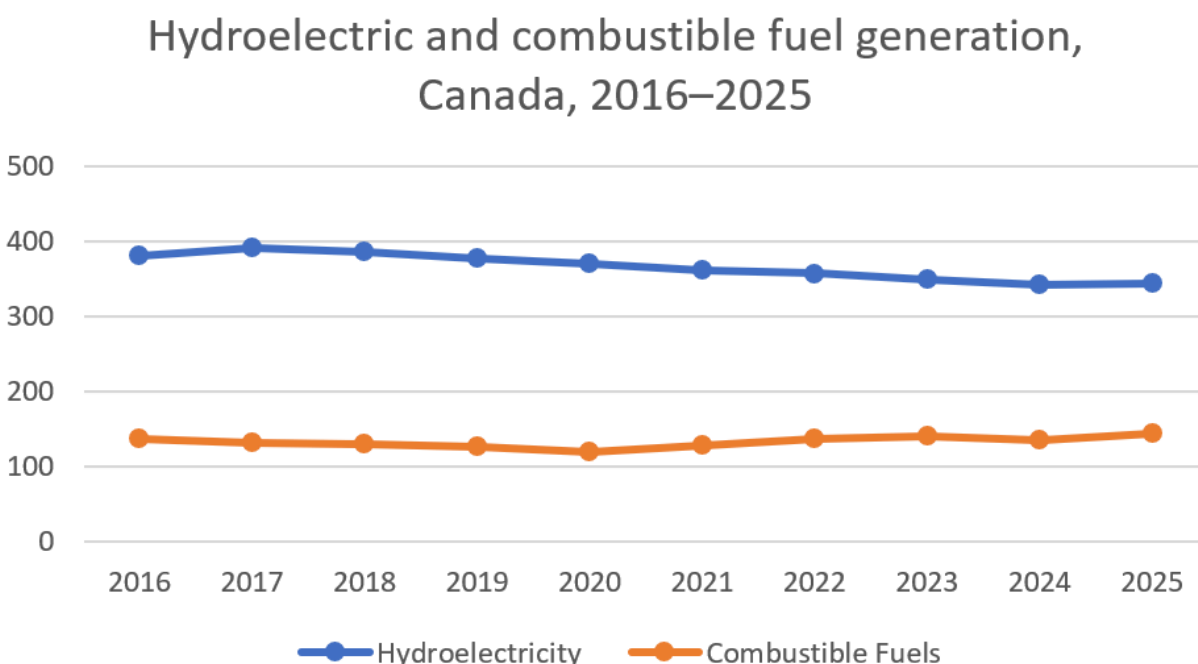
Canada's electricity sector has achieved several notable gains in its quest toward sustainability. Since 2005, greenhouse gas emissions from electricity generation have fallen by 58%, which is the single largest sectoral reduction in the Canadian economy over that period.¹ The grid that powers Canadian homes and industries is, in a normal year, over 80% non-emitting.² Historically, hydropower accounted for nearly 60% of the power grid.³ In individual provinces like British Columbia and Manitoba, the contribution of hydro generation reaches 90% and beyond.⁴

However, the past few years have shown why this dependence on hydro poses a vulnerability. The years 2023 and 2025 stand out in particular, both featuring droughts and some of the lowest hydro generation levels since Statistics Canada's current series began in 2016. Conversely, combustible fuel generation rose to its highest level since 2016 in 2025. Canada became a net electricity importer during multiple months of 2024 and 2025, the first time this had occurred in the current era.⁵ This paper examines the pattern indicated by the data and finds that it reflects a structural risk: Canada's electricity system is highly reliant on hydro generation for meeting its clean energy goals, but hydro's variability and the absence of clean fallback solutions can contribute to increased reliance on fossil fuel generation. As the net-zero transition continues, accounting for variations in hydro generation while also diversifying beyond hydro can make the grid more reliable in delivering clean energy even when under pressure.

The Structural Risk

The scale of the shift in the last few years is measurable. Canada's hydroelectric generation peaked at 392 TWh in 2017. By 2024 it had fallen to 341.8 TWh: a drop of roughly 50 TWh from that peak.⁶ Expressed as a share of total generation, hydro fell from 61% in 2022 to just under 55% in 2025, a series low. At the same time, combustible fuels moved in the opposite direction, rising from 137.8 TWh in 2022 to a series-high 143.4 TWh in 2025.⁶

A simple way to illustrate the scale of this vulnerability is to translate hydro variability into a supply shock. Using 2024 hydro generation of 341.8 TWh, a 10% decline would imply a shortfall of roughly 34.2 TWh.⁶ That is about one-quarter of 2025 combustible-fuels generation (143.4 TWh), meaning even a moderate drought can create a gap large enough to require substantial replacement, which is currently often met by fossil generation or higher-emissions imports.



All values are in Terawatt-hours (TWh).

Source: Statistics Canada, Table 25-10-0015-01

British Columbia illustrates one version of this dynamic. The province generates the vast majority of its electricity from hydroelectric dams, with almost no domestic fossil backup. When Williston Reservoir inflows hit record lows, BC Hydro faced a generation deficit of 10,430 GWh and covered it by importing close to 25% of provincial electricity demand at a cost of approximately \$1.38 billion.⁷ A large share of those imports came from the United States and

Alberta, whose grid ran at approximately 74.7% natural gas in 2024, with a carbon intensity more than four times the Canadian national average.⁸ The result is a meaningful fossil footprint that is not reflected in provincial generation-based emissions accounting. Provincial emissions inventories count only what is generated within a province's borders, so the carbon embedded in those imports does not appear in BC's official figures. The \$1.38 billion paid for imports is the financial record of a shortfall that likely increased reliance on fossil fuel generation outside the province.

Manitoba provides a more important example by showing how current decisions are reinforcing this systemic dynamic for the future. The province generates around 97% of its electricity from hydro.⁹ This presently makes it one of the cleanest grids in Canada, but the concentration of hydro also means there is no domestic clean alternative when precipitation falls short. In response, Manitoba Hydro has proposed a \$3 billion combustion turbine facility at Brandon, designed specifically to provide dispatchable backup when hydro is insufficient, with natural gas as the lowest-cost fuel option under consideration.¹⁰ The utility's own planning documents describe this dispatchable capacity as "essential" to meeting demand through the next decade.¹¹

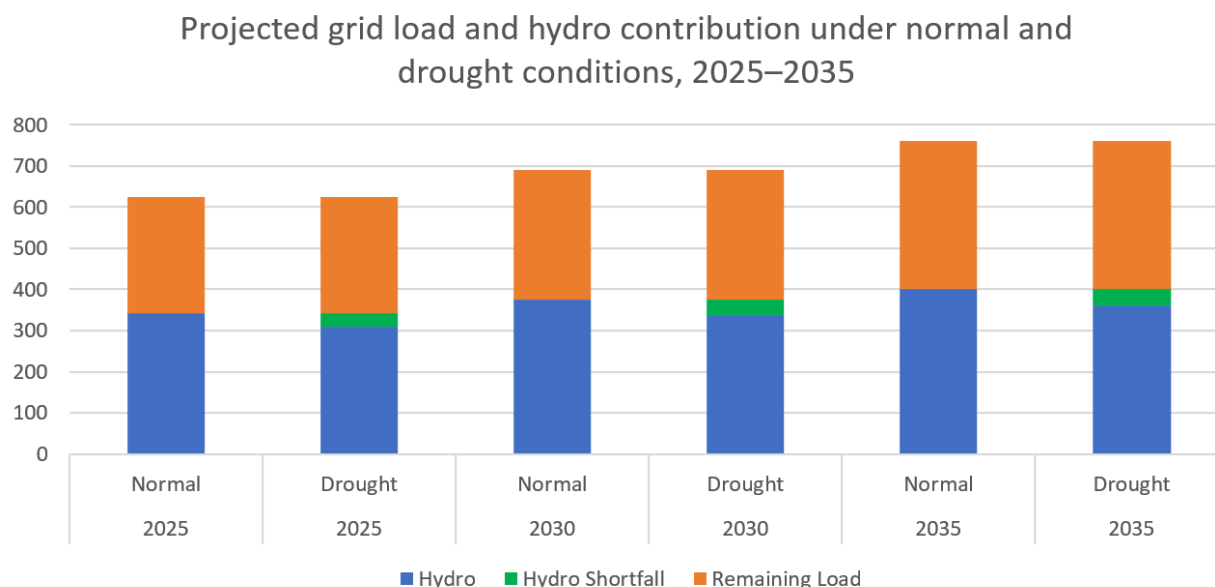
This measure is understandable today given that clean and renewable sources, while rapidly growing, still pose practical issues regarding costs and reliability. But this measure also reveals why clean alternatives must be proactively identified and developed. By turning to fossil fuels to cover the future shortfalls of hydro, the province starts creating a system that, when under pressure, leads to high emissions under stress conditions. Such design decisions sustain the presence of fossil fuels, making them recurrent players because the irregularity associated with hydro generation is a recurring issue.

Electrification and the Compounding Risk

The risks outlined above are compounded by Canada's broader push toward electrification. Electrification is a key component of the federal government's environmental strategy. The Clean Electricity Regulations, finalized in December 2024, establish emission limits on electricity generating units beginning in 2035, with a net-zero grid target of 2050.¹² Running alongside this is a plan to move transportation, space heating, and industrial processes off fossil fuels and onto electricity, which will require substantially more electricity than Canada currently produces. The Canada Energy Regulator (CER) projects that electricity demand will grow between 26% and 44% by 2050 relative to 2023 levels.¹³ BC Hydro's own filings project a 15% increase in provincial electricity demand by 2030 alone, driven by electric vehicles, population growth, and restrictions on natural gas for heating.¹⁴

The problem is that this rising demand will land on a grid whose clean foundation has shown vulnerability under stress conditions. The below-average precipitation years demonstrated that a 10% drop in hydro output – consistent with what Statistics Canada recorded nationally in 2023 – leaves a gap of tens of terawatt-hours that the current system fills largely with fossil fuels.¹⁵ CER

scenario projections show that electricity demand will rise faster than hydro generation, which would make the grid more sensitive to future hydro declines simply because the baseline requirements are higher.^{13,16} As total grid load grows with electrification, droughts of the same intensity can create a greater shortfall and require an earlier intervention from other sources. As it stands today, these other sources largely consist of fossil fuels. In short, while electrification remains essential for decarbonization overall, without parallel expansion of reliable clean capacity, it may increase reliance on fossil fuels during supply stress.



All values are in Terawatt-hours (TWh). Each year shows a normal-year bar (left) and a drought-year bar (right).

2025 actuals: Statistics Canada, Table 25-10-0015-01.

Projections are illustrative, not forecasts. 2030 and 2035 values interpolated from CER Canada's Energy Future 2023, Current Measures scenario.

Drought scenario applies a 10% hydro decline, consistent with the 9.3% national decline recorded in 2023 (Statistics Canada).

Possible Solutions

While hydroelectricity has a crucial role to play, a lack of attention has been given to clean sources that replace hydro when it underperforms. To be sure, fossil fuels are an appropriate solution in some cases, but presently they are a common fallback solution. Besides expanding the storage capacity of wind and solar and funding clean yet relatively undeveloped sources like geothermal energy,¹⁷ there are two other key steps that can be begun immediately to resolve this issue.

The first concerns geographic fragmentation. Hydro conditions across Canada are not uniform. A drought affecting British Columbia's reservoirs may not affect Quebec's or Manitoba's to the same extent in the same year. Moreover, provinces like New Brunswick and Ontario (with its nuclear plants) can supply clean energy from sources other than hydro. A grid with sufficient interprovincial transmission capacity could allow clean surplus generation in one region to cover

deficits in another, reducing the need for provinces to turn to fossil fuels when its own hydro underperforms. At present, Canada operates three largely separate regional grids – Western, Eastern, and Québec – with limited east-west interconnection, and provinces trade more electricity with US states than with each other.¹⁹ Modelling by the Canada Electricity Advisory Council and the Canadian Climate Institute suggests that expanded interprovincial interties could reduce system-wide emissions and lower industrial electricity prices, while also increasing resilience to precisely the kind of regional supply stress seen in recent drought years.²⁰

Energy policy largely falls under the jurisdiction of provinces, and so implementing the first measure would require addressing two key provincial concerns: cost burdens and a loss of flexibility.²² Because transmission investments are evaluated primarily on provincial cost-benefit grounds – even when their largest gains are national – provinces have limited incentive to build interties on their own, which makes federal financing and coordination essential. At the same time, there must be room for provincial governments to make tailored contracts that align with their specific economic and environmental goals.

The second response does not require new infrastructure but a change in planning assumptions. Utilities and regulators have often treated hydro as highly reliable firm capacity, but greater attention must be given to hydrological variability. Under that assumption, utilities and regulators have had little reason to require non-fossil backup capacity for drought scenarios, since hydro was not expected to fail. The 2023–2025 data challenge that assumption, and the Clean Electricity Regulations have begun to reflect it: one of the regulatory flexibilities explicitly included is the ability for electricity units to bank compliance credits for use in drought years when hydro falls short.²¹ If grid planning frameworks followed the same logic and accounted for hydro's variability in drought scenarios, the consequence can be direct: utilities would be required to secure non-fossil backup for periods when hydro underperforms, rather than defaulting to the cheapest dispatchable generation when a drought arrives. That is the condition under which fossil fuels currently fill the gap, and changing the planning assumption can meaningfully change it.

Conclusion

Canada's electricity grid is cleaner than most, and the progress made since 2005 is substantial. Recent years, however, indicate that this performance is partly contingent on hydroelectric availability. Periods of reduced precipitation have coincided with increased fossil fuel generation and imports, consistent with a grid that lacks sufficient clean alternatives when hydro underperforms. British Columbia covered a quarter of its demand through high-carbon imports during low-water conditions, while Manitoba has proposed new dispatchable gas capacity to address similar risks going forward. These are not isolated decisions but expressions of an underlying design gap that should be addressed as the transition to clean energy continues. Electrification is likely to place greater pressure on that gap by raising the overall load that must be met when hydro falls short. Addressing it will require expanding non-emitting generation

capacity, strengthening interprovincial transmission so that clean surpluses in one basin can cover deficits in another, and revising the planning assumption that hydro will always deliver.

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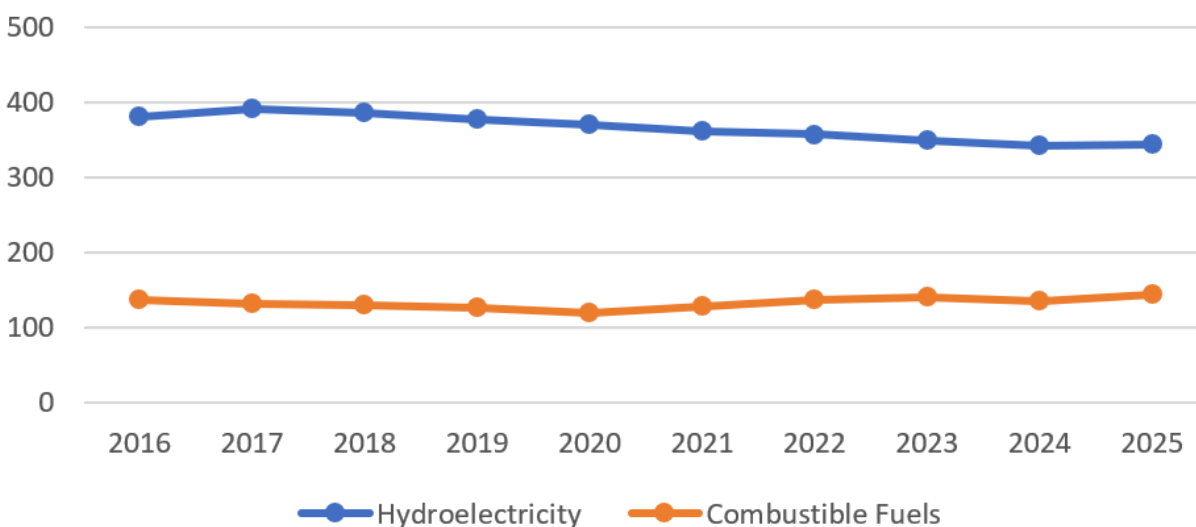
The Hydro-Emissions Tension: Addressing Structural Vulnerabilities in Canada's Net-Zero Transition

Executive Summary

Canada's electricity system is among the cleanest in the world, largely because of its heavy reliance on hydroelectricity.¹ But recent droughts have revealed a structural vulnerability: when hydro output falls, provinces often turn to increased fossil fuel generation or high-emitting imports to fill the gap.^{2,3} This pattern is already visible in British Columbia and Manitoba, and it risks becoming more pronounced as electrification drives electricity demand higher.⁴

This brief argues that to effectively secure Canada's net-zero electricity transition, governments and utilities do three things: expand clean dispatchable backup capacity, strengthen interprovincial transmission, and revise planning assumptions to reflect hydro's constraints and exposure to variability. Without these changes, shortfalls in hydro generation might continue to push emissions higher during periods of supply stress.^{3,5} As the net-zero transition continues, accounting for variations in hydro generation while also diversifying beyond hydro can make the grid more reliable in delivering clean energy even when under pressure.

Hydroelectric and combustible fuel generation, Canada, 2016–2025



All values are in Terawatt-hours (TWh).

Source: Statistics Canada, Table 25-10-0015-01

Policy Recommendations

Federal and provincial decision-makers should align electricity planning with hydro variability by:

- Expanding non-emitting backup capacity – including storage, geothermal, and other firm low-carbon resources⁶
- Building stronger interprovincial transmission links so clean electricity can move across regions when local hydro basins are stressed⁷
- Updating planning and regulatory assumptions to require utilities to prepare for drought years without defaulting to fossil fuels⁸

Problem Statement

Canada has made real progress decarbonizing its electricity sector. Since 2005, greenhouse gas emissions from electricity generation have fallen sharply, and the grid now runs over 80% non-emitting in a normal year.^{9, 10} Much of that comes from hydroelectricity, which supplies nearly 60% of national electricity – and in provinces like British Columbia and Manitoba, 90% or more.^{1, 11}

The problem is that hydro is not immune to climate stress. In recent years, below-normal precipitation and drought have reduced hydro output across Canada. In 2025, combustible fuel generation reached its highest level since 2016, while hydro's share of total generation fell to a series low, which together highlight growing system stress. Canada also became a net electricity importer during multiple months in 2024 and 2025 – the first time this had happened in the current era.^{2, 3} These trends may not be one-off anomalies because the grid is built around hydro abundance and has limited clean options when that abundance runs short.

The core problem is not hydro itself. It is that Canada has not developed enough clean backup capacity to replace hydro when it underperforms. When that gap opens, fossil fuels and imports become the default.

The Structural Risk

The scale of the vulnerability is measurable. Canada's hydroelectric generation peaked at 392 TWh in 2017 and fell to 341.8 TWh in 2024. A 10% decline from that level would create a shortfall of roughly 34.2 TWh – about one-quarter of 2025 combustible-fuels generation. Even moderate droughts can create supply shortages large enough to require substantial fossil replacement.³

British Columbia illustrates the problem clearly. During low-water conditions, BC Hydro covered close to a quarter of provincial electricity demand through imports, at a cost of about \$1.38 billion.¹² Much of that imported electricity came from Alberta and the United States, where generation is substantially more carbon-intensive than in Canada's hydro-heavy provinces.¹³ Because emissions inventories count only in-province generation, the emissions associated with those imports are not fully visible in provincial totals.

Manitoba shows how this problem can become locked into future planning. The province relies on hydro for about 97% of its generation, leaving little domestic clean backup when precipitation is low.¹⁴ Manitoba Hydro has proposed a \$3 billion combustion turbine facility at Brandon to provide dispatchable backup, with natural gas as the lowest-cost fuel option under

consideration.¹⁵ The project is described in utility planning documents as essential for meeting demand through the next decade.¹⁶ Since hydro shortfalls are a recurring risk, dependence on fossil fuels to compensate for them risks turning what should be an emergency measure into a structural feature of the grid.

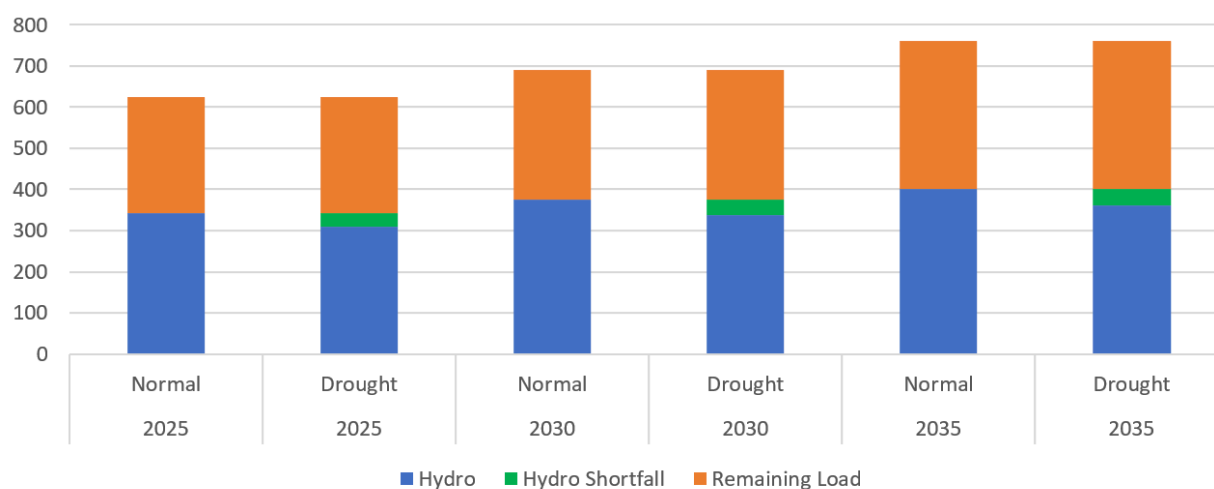
Why Electrification Makes the Problem More Urgent

Canada's electrification push is the right long-term direction. The Clean Electricity Regulations, finalized in December 2024, set emission limits on generating units beginning in 2035, with a net-zero grid target of 2050.¹⁷ The Canada Energy Regulator projects electricity demand growth of 26% to 44% by 2050, and BC Hydro expects significant provincial demand increases by 2030 driven by electric vehicles, heating electrification, and population growth.^{4,18}

The concern is not the direction – it's the timing and the gap. Rising demand increases the cost of hydro shortfalls. A drought that caused manageable disruption in the past may require far larger amounts of replacement generation as total load grows.^{4,19} If that replacement still comes mainly from fossil fuels, electrification can unintentionally push emissions up during stress periods even as it reduces them over the long run.

The core risk is not that electrification is the wrong strategy. The risk is that electrification is being pursued faster than the clean system needed to support it. Without parallel investment in clean backup and grid integration, Canada may become more dependent on fossil fuels when hydro generation weakens.

Projected grid load and hydro contribution under normal and drought conditions, 2025–2035



All values are in Terawatt-hours (TWh).

Each year shows a normal-year bar (left) and a drought-year bar (right).

2025 actuals: Statistics Canada, Table 25-10-0015-01.

Projections are illustrative, not forecasts. 2030 and 2035 values interpolated from CER Canada's Energy Future 2023, Current Measures scenario.

Drought scenario applies a 10% hydro decline, consistent with the 9.3% national decline recorded in 2023 (Statistics Canada).

Policy Options

Three responses address different parts of the same design gap.

Expand clean dispatchable capacity

When hydro output is low, Canada needs more non-emitting resources capable of running battery storage, such as geothermal energy, and other firm low-carbon sources.⁶ These can fill the gap that wind and solar, without substantial storage, currently struggle to cover reliably across multi-month drought periods. Despite its greater reliability, Canada currently generates less than six megawatts of geothermal electricity nationally, and only three provinces have geothermal-specific legislation, with no national strategy in place.²¹

Strengthen interprovincial transmission

Hydro conditions across Canada are not uniform. A drought affecting British Columbia's reservoirs may not affect Quebec's or Manitoba's to the same extent in the same year. Moreover, provinces like New Brunswick and Ontario (with its nuclear plants) can supply clean energy from other sources. Stronger east-west transmission links would allow provinces with surplus clean electricity to support those running short, reducing dependence on fossil imports and making the national grid more resilient. At present, Canada's three regional grids operate largely in isolation from each other, and provinces typically trade more electricity with U.S. states than with one another.⁷

Because electricity policy is largely provincial, intertie development is constrained by cost allocation and flexibility concerns. Since the benefits of transmission are often national while costs are local, progress will likely require federal financing alongside room for provinces to make tailored contracts that preserve control over how electricity is used and traded.²⁰

Reform planning assumptions

Utilities and regulators have often treated hydro as highly reliable firm capacity, but greater attention must be given to hydrological variability. Planning frameworks should explicitly model hydro variability and require non-fossil backup for drought scenarios. The Clean Electricity Regulations have already moved in this direction: they include a provision allowing electricity units to bank compliance credits for use in drought years when hydro falls short.⁸ The next step is for grid planning – not just emissions compliance – to reflect the same logic. If utilities were required to plan for drought-year hydro shortfalls, they would have a strong incentive to secure clean backup capacity in advance, rather than reaching for whatever dispatchable generation is available when a drought arrives.

Conclusion

Canada's grid is cleaner than most, but its low-emissions record depends heavily on hydro availability. When hydro underperforms, the system often falls back on fossil generation and imports. British Columbia and Manitoba illustrate that this isn't a temporary anomaly – it's a structural design problem, one that electrification will make more consequential as overall electricity demand grows.

The answer is not to slow electrification. It is to build the grid capable of supporting it under stress. That means clean backup capacity that doesn't depend on rain, transmission infrastructure that lets clean electricity move to where it's needed, and planning rules that stop treating hydro as a guaranteed resource. Without such changes, Canada's electricity system risks remaining clean in good years and more fossil-dependent when conditions turn difficult.

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